

Welcome to Kickstart

Welcome to Computer Science and the programming kickstart course! The exercises below will quickly get you up and running with programming P5 JavaScript. We have intentionally left some things unexplained as we want you to want you to experiment on your own.

Resources

There is a world of information and resources to help you learn p5.js. Check out the following:

Book McCarthy, Lauren, Casey Reas, and Ben Fry. Getting started with P5. js: Making interactive graphics in JavaScript and processing. Maker Media, Inc., 2015.

p5.js Web site The official website Click to go to the URL

Happy Coding A fun set of tutorials and examples for p5.js Happy Coding. Click

Kickstart Absalon You can find the book and other materials on the LMS.

Have Fun



First Program

Type the following example in the Processing editor and press -button:

```
function setup() {  
  //set the canvas and background color  
  createCanvas(400, 400);  
  background(220);  
  
  // Tree  
  rect(55, 50, 10, 20);  
  ellipse(60, 35, 30, 40);  
}  
  
function draw() {}
```

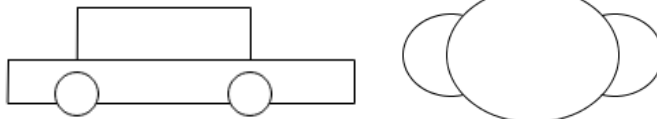
Then add:

```
// powerplant  
rect(120, 50, 60, 30);  
rect(160, 20, 10, 30);  
triangle(120, 50, 136, 40, 136, 50);  
triangle(136, 50, 152, 40, 152, 50);
```

And:

```
# Vindmølle  
// windmill  
line(300, 50, 320, 51);  
line(300, 50, 289, 67);  
line(300, 50, 291, 32);  
line(300, 50, 300, 90);
```

Now try drawing a car and a cloud:



Colors

You are now going to colour the shapes. To do this, use the `fill(r, g, b)` function, which selects which colour to use for fill and text colour. It's all about calling `fill` in the right places! As arguments, you specify the amount of red (0-255), blue (0-255) and green (0-255).

Her er nogle grundfarver:

```
fill(255, 0, 0); // red          fill(0, 0, 0); // black
fill(0, 255, 0); // green        fill(255, 255, 255); // white
fill(0, 0, 255); // blue         fill(255, 255, 0); // yellow
```

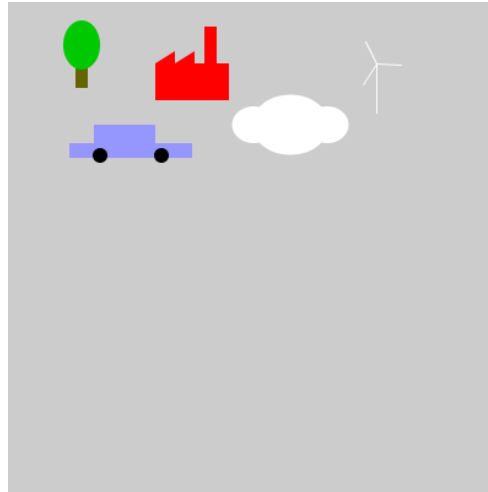
Optionally, find colours using an online color picker or RGB colour table. For example, search for “rgb color picker”.

Lines and outlines

To specify the colour of strokes (e.g. `line`) and outlines, use `stroke(r, g, b)`. Also try the `noStroke()` function, to turn off outline drawing.

Example

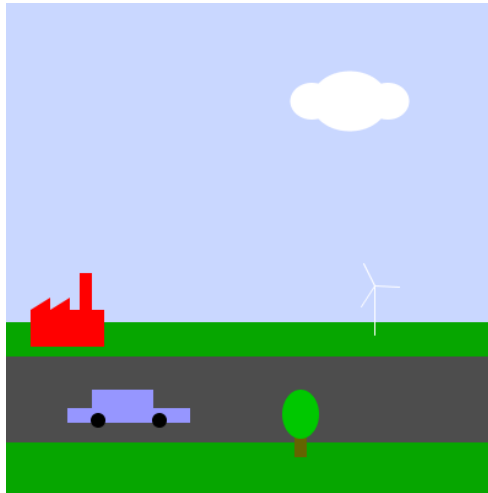
Here's an example of what it might look like after colouring:



Green City

Use what you've learned to make it a slightly nicer scene, with background, foreground and the extra details you think should be there. For example, I've drawn a road and moved the characters around.

Tip: To change the background color from gray you can use `rect` which you already know, but you can also use the `background(r, g, b)` command. It will delete everything and fill the screen with the specified colour.



Remember to use comments to find your way around the code easily! Commenting code is good because it improves code readability and maintainability by explaining the purpose and functionality of the code to others and to your future self.

Variables

Create a new project (“File” -> “New”) and immediately save it the project. Call it “Aquarium”.

Type in this piece of code:

```
function setup() {  
  createCanvas(400, 400);  
  background(220);  
  
  //basic fish shape  
  fishX = 150;  
  ellipse(fishX, 200, 120, 75);  
  triangle(fishX - 60, 200, fishX - 90, 170, fishX - 90, 230);  
}  
  
function draw() {  
  
}
```

Try changing 150 to a different number in the `fishX` specification.

Now add the following:

```
eyeSize = 15;  
ellipse(fishX + 30, 190, eyeSize, eyeSize);
```

Try changing the value of `eyeSize`.

Tasks:

You have now added a fish that can be moved just by changing one value.

- Color the fish
- Give the fish a fin that moves when you change `fishX`.
- Give the fish a pupil that moves when you change `fishX`.
- Create a new variable, `fishY`, that controls the y-position of the fish

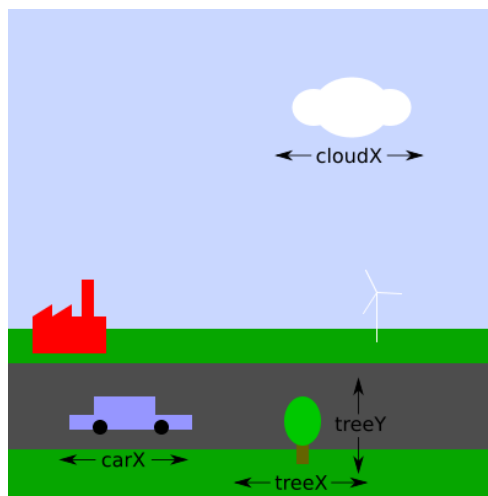


Remember to save the project. We'll be working on it later.

Green City continues

Switch to the electric car project and introduce variables to specify the location of the objects so that we can later animate these objects.

- Create a variable `carX` so the car can move forward and back
- Create a variable `cloudX` so the cloud can move back and forth
- Create a variable `treeX` so the tree can be moved horizontally
- Create a variable `treeY` so the tree can be moved vertically



Aquarium Continues

Use what you've learned to expand your aquarium project; here's an example, but feel free to use your imagination! For example, one variable is used for the x-coordinate of the seaweed plant and another variable for the x-coordinate of the entire group of rocks (as a unit). An additional set of variables `fish2X/fish2Y` to control the location of the additional fish.

